Resonant cascaded Kerr nonlinearity in $\chi^{(2)}$ metasurfaces: supplemental document

Derivation of the cascaded susceptibility of a resonator, its uniform-medium limit, the bandwidth sum rule and the thresholds, the host-material parameters, the coupled-mode simulations, the singly resonant metasurface, and the microring model.

1. MODAL PROJECTION AND COUPLING CONSTANTS

Projection of the wave equation with dispersive permittivity onto a mode $\mathbf{f}_n$ with slowly varying amplitude a_n [1],

$$\dot{a}_n = i(\Omega - \omega_n)a_n + \frac{i\Omega}{2N_n}\int \mathbf{f}_n^* \cdot \mathbf{P}\, d^3r,$$

with $N_n = \frac{1}{2}\int(\varepsilon_0\tilde{\varepsilon}_n|\mathbf{f}_n|^2 + \mu_0|\mathbf{h}_n|^2)\, d^3r$ and $\tilde{\varepsilon}_n = \partial(\omega\varepsilon_r)/\partial\omega$ at ω_n. Applied to the second-harmonic, difference-frequency and self-phase-modulation sources, it gives the couplings

$$\beta = \frac{\varepsilon_0\omega_1 M}{N_2}, \qquad \alpha_3 = \frac{3\varepsilon_0\omega_1}{2N_1}\int \chi^{(3)}_{ijkl} f^*_{1,i} f_{1,j} f_{1,k} f^*_{1,l}\, d^3r,$$

the Manley–Rowe relations, and the stored energy $2N_n|a_n|^2$.

2. QUASINORMAL-MODE GREEN'S FUNCTION AND THE CASCADED KERNEL

Pole expansion of the Green's function at $2\omega_1$ for a leaky second-harmonic mode [2, 3], the nonlocal kernel $\chi^{(2)} \cdot G \cdot \chi^{(2)}$, and its reduction to Eq. (1) of the main text when the pump is a single mode. The same equation follows from adiabatic elimination of the second-harmonic amplitude in the coupled-mode equations, including the full Lorentzian dependence on $\Delta\omega$.

3. UNIFORM-MEDIUM LIMIT AND A LAYERED-CELL TEST

Bound second-harmonic wave in a uniform medium, its longitudinal part, and the reduction of Eq. (1) of the main text to the bulk susceptibility for $\eta_{\mathrm{geom}} = 1$. Comparison of the single-mode result with the full driven field of a one-dimensional layered Bloch cell (Fig. S1).

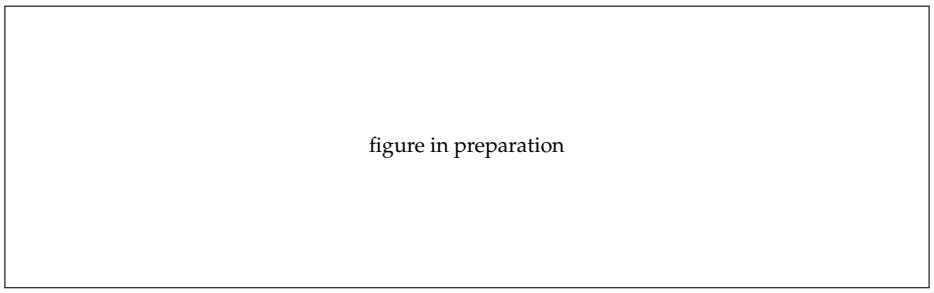


Fig. S1. (a) Kerr coefficient of the pump mode from integration of the two-mode equations (markers) and from the closed form (line) versus $\Delta\omega/\gamma_2$. (b) Second-harmonic amplitude from the full driven field of a layered Bloch cell divided by the single-mode prediction, versus $\Delta\omega/2\omega_1$.

4. SUM RULE, LOSS AND THRESHOLDS

Response of the cascade to a modulated pump, the peak and width of the cascaded coefficient, and the Q_2-independent product of Sec. 3 of the main text. Loss-to-Kerr ratio, the conditions for bistability and parametric gain, and the onset of parametric instability at $|\beta a_1| \approx |\Delta\omega|$ [4]. Comparison with phase-mismatched bulk cascading [5].

5. HOST MATERIALS

Material constants and the resulting enhancement ratios (Table S1).

Table S1. Cascaded Enhancement at $Q_2 = 4000$, $\Delta\omega = 2\gamma_2$ and $\eta_{\mathrm{geom}} = 0.3$

Host	$\lvert\chi^{(2)}\rvert$ (pm/V)	$\chi^{(3)}_{\mathrm{nat}}$ (m^2/V^2)	n_2	R	Q_2 at $R = 1$
$LiNbO_3$	50	2×10^{-21}	2.26	49	82
GaP	100	1×10^{-19}	3.1	2.1	1.9×10^3
AlN	10	2×10^{-21}	2.1	2.3	1.8×10^3
GaAs	350	1.4×10^{-18}	3.5	1.4	2.8×10^3
AlGaAs	100	1×10^{-18}	3.3	0.18	2.2×10^4

6. COUPLED-MODE SIMULATIONS

Equations, integrator, tolerances, run lengths and convergence criterion for Figs. 3 and 4 of the main text and Fig. S1, and the time-domain model for pulsed excitation used in Fig. 4(d) of the main text.

7. SINGLY RESONANT METASURFACE

Definition of $\widetilde{M}$, its extraction from full-wave fields, the unit-cell average, the parameters of the design, and the Z-scan model.

8. MICRORING

Photon-number form of the coupled-mode equations and the relation between the coupling rate and the normalised second-harmonic efficiency [6, 7]; the bistability threshold at the pump-frame mismatch; the multimode stability analysis for the parametric threshold; the thermal and photorefractive model [8]; and the electro-optic tuning of $\Delta\omega$.

9. PARAMETERS

Material constants, mode parameters and design points used in all figures and tables.

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